PREDICTING THE INFLUENCE OF INTERNET CULTURE ON KALSHI SPENDING: AN LLM AGENT APPROACH TO WHAT WILL SHAQUILLE O'NEAL SAY DURING INSIDE THE NBA?

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ABSTRACT

Prediction markets increasingly list betting lines influenced by the internet and meme culture alongside traditional politics, financial, and sports markets. This research examines whether a market's genre of betting, influences the trading volume it attracts on Kalshi. Using an LLM built from the open source tool Langflow, and powered by the open source AI, "LLama," 1,433 betting lines were categorized as "Meme," "Serious," and "Grey." These categories were determined by human feedback, theory, and real world events all through the LLM. These categories were also further labeled "sports" and "non-sports." Ordinary least squares regression was used to model trading volume as a function of category, bet type, and market duration (days). Relative to meme labeled markets, serious markets were associated with significantly higher volume ($p < .001$ for Models 2 and 3). Sports bets were related to less volume compared to non sports bets across the models, and longer market duration was associated with higher volume. These preliminary results indicate that markets tied to consequential real world events attract more capital than meme driven markets.

Keywords First keyword · Second keyword · More

1 Introduction

Prediction markets grow year after year as the varying markets to bet on expand. One of the main selling points, is the ability to bet on absurdity, memes, and implausibility. Prediction markets include bets on predicting what Trump will say on his next speech, if GTA VI will ever release, and if we can expect a thunderstorm tomorrow. Little is known whether internet culture markets command real trading volume comparable to more conventional, traditional markets, or whether their popularity is mostly noise with minimal capital behind it.

This paper sets out to answer that question using an original data set of 1,433 Kalshi betting lines, classified by an LLM and sorted into three categories "Meme," "Serious," and "Grey." Each category was further evaluated for whether it constituted a sports bet, since sports make up the largest single category of listings, and also the control variable of duration for how long the betting line stayed open on the market. Using this classification, the paper tests whether a market's cultural framing predicts the amount of volume it attracts, and whether that relationship changes once sports and non-sports markets are considered separately and market duration.

2 Methods

A brief overview of the methods used throughout the research conducted include the process of building an LLM to process betting lines, and output into the three categories of "Meme," "Serious," and "Grey." The data set used for this research is from huggingface by the author Deniz Zorlu[1] and includes 1,433 rows of betting lines, dates, tickers, and more.

[1]Zorlu, D. kalshi-filtered. Hugging Face. Retrieved from dzorlu/kalshi-filtered.

2.1 Category Definitions

To keep the LLM's classification consistent, each category had specific defined criteria the agent followed per each betting line.

Meme markets were defined as bets tied to pop and internet culture, celebrity behaviors, unfathomable events, or events with no real world or political consequence (e.g., celebrity breakups, Twitter/X drama, joke crypto tokens, religious internet memes, crazy ideas). Betting on these primarily represents signaling involvement with pop-cultural discourse; earnest or ironically.

Serious markets were defined as bets tied to real world financial and political events, Standard political, macroeconomic, geopolitical, or highly consequential real-world events (e.g., presidential elections, central bank rates, major legislation, sport questions with clear real stakes like a golden boot winner). Betting on these primarily represents an individual using knowledge to make money.

Grey markets that are generally niche, unusual, uncommon, or technically complex markets that don't cleanly fit Serious or Meme (e.g., obscure scientific replication studies, minor localized weather data, hypothetical sports scenarios). Betting on these could equally reflect hyper specific knowledge or an irregular bet.

2.2 Creating the LLM agent

To start the data collection process, an LLM agent was necessary in sorting the data in an efficient manner. Using the app Langflow[2], which is a free agent building service, and an API key[3] for the ai model llama-3.1-8b-instant[4],for the ai model llama-3.1-8b-instant, the first variation of meorser (the category sorting agent, shortened for "MemeorSerious") was built. The first variation was unique in the way that it broke down each betting line you fed it and provided an in depth reason as to why said betting fell into each category. In order to ensure more accurate results, I fed it 50+ markets and had it label them to what category fit best. I included examples of what to do, and what not to do in my prompt. Through vibe based prompting and constant adjustments, eventually a consistent working agent was formed. After accurately representing what each category the various markets fell into, the next variation was created. The second variation of meorser, reads large .csv files and outputs them into a neatly organized data table that additionally includes the category for each betting line, and also the classification of sports and non-sports.

2.3 Data Collection

For data collection, the dataset was processed through meorser and the output contained the added columns of category and sports/non-sports. The results are shown in the two histograms below. The first histogram represents the % of each category in the dataset and the second histogram represents the average volume per each category.

[2] Langflow. (2026). Langflow: An open-source visual framework for building LLM applications. `https://www.langflow.org`.
[3] API access provided via GroqCloud. `https://groq.com`.
[4] Meta AI. (2024). Llama 3.1 (8B Instant). `https://ai.meta.com/llama/`.